

Unnamed_Unit

Unit 1 - Exploring the Core

PDF Documents

PDF Document 1: 1738068072108-Chapter 7_unit 1 - Exploring the Core.pdf

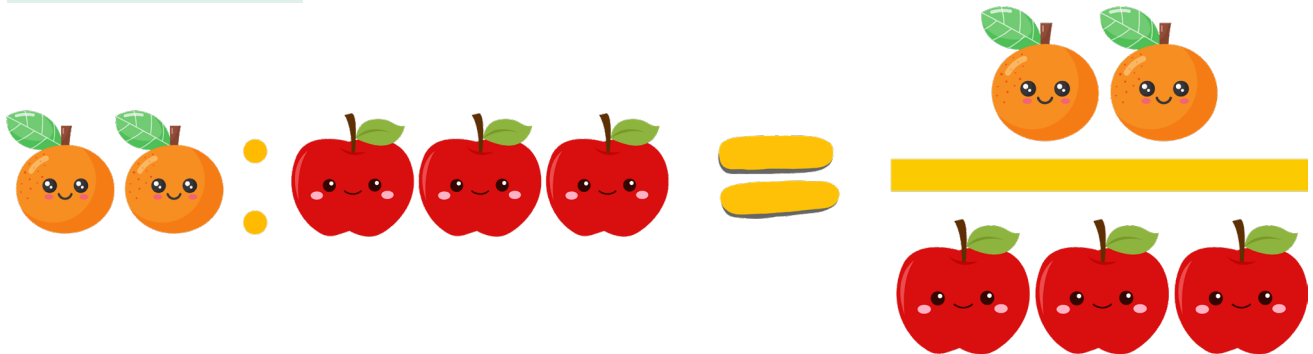
This PDF document is included in the unit materials.



UNIT 1 EXPLORING THE CORE

Topic Investigation

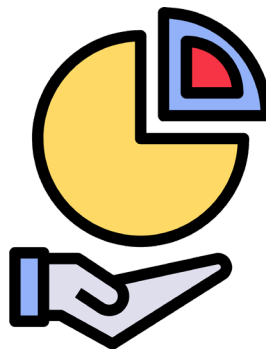
WHAT IS A RATIO?



Simply put, a ratio tells us how many times one value is contained within another. For instance, if there are 2 oranges for every 3 apples, we can express this relationship as a ratio of 2:3. Ratios can be written in three ways: as a fraction ($\frac{2}{3}$), with a colon (2:3), or using the word "to" (2 to 3).

VISUALIZING RATIOS

To help you better understand ratios, we'll start with some interactive activities. You'll use colored blocks to create visual representations of different ratios. For example, if given a ratio of 4:1, you would use four blocks of one color and one block of another. This hands-on approach makes it easier to see how ratios work.



RATIOS IN DAILY LIFE

Ratios are everywhere! Whether you are cooking and need to mix 2 cups of water with 1 cup of rice or you're trying to figure out if you have enough paint to cover a wall, understanding ratios is key. We'll explore some common scenarios where ratios are used and practice setting up and solving ratio problems based on these real-life situations.



THE IMPORTANCE OF RATIOS IN MATHEMATICS

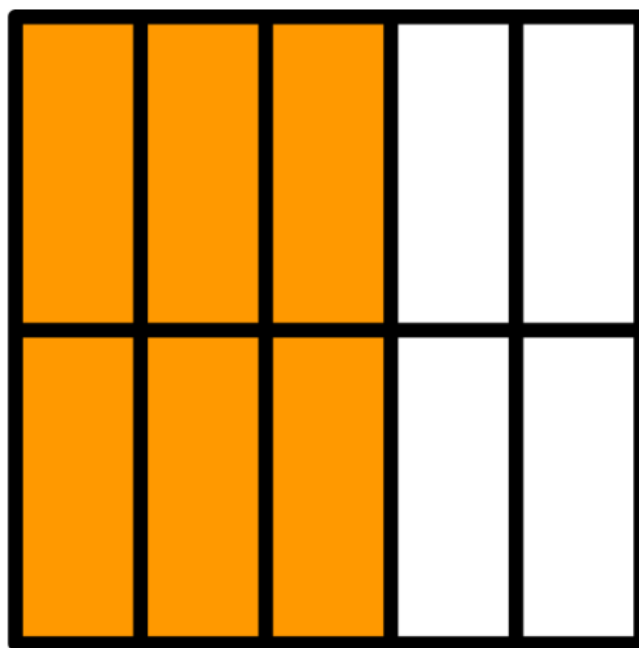
Ratios are not only practical but also a cornerstone in many areas of mathematics and science, especially in proportions, rates, and even algebra. Understanding ratios will help you grasp more complex concepts more easily as they form the basis for much of the math you will encounter in the future.

$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$$



SUMMARY

By the end of this unit, you will have a solid understanding of what ratios are and how to use them effectively. You'll be able to identify ratios around you and apply this knowledge in practical situations. Get ready to look at the world a little differently—through the lens of ratios!



TIPS

Understanding ratios is very important in our everyday lives. Without a grasp of ratios, we could face many disadvantages.



Inquiry & Reflection

Discussion question 1

If a recipe uses a ratio of 2 parts juice to 5 parts water and you have 10 cups of juice, how many cups of water do you need? Why do ratios matter in recipes?

Discussion question 2

You have 12 cookies to share equally with 3 friends. What ratio would you use to divide the cookies? Why is it important to use ratios fairly?